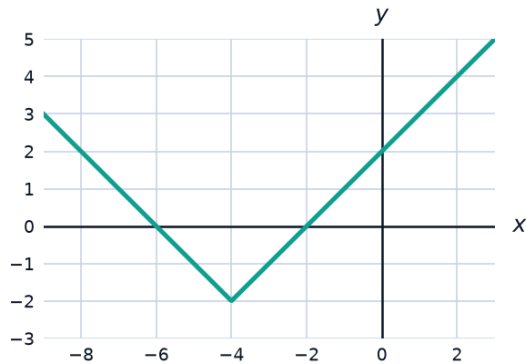


Functions: Modulus & Inverse



Evaluating and graphing modulus functions

- 1 Given $f(x) = |2x - 3|$, find $f(1)$ and $f(-2)$.
- 2 The graph shows $g(x) = |x + 4| - 2$. State the coordinates of its vertex and the range of g .



Solving modulus equations

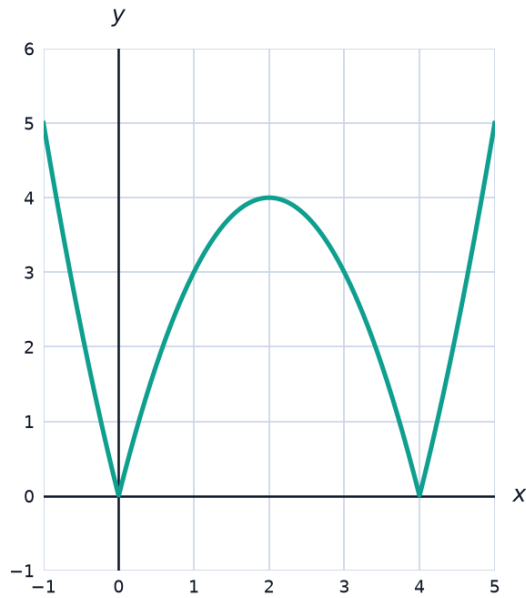
- 3 Solve $|3x - 2| = 7$.
- 4 Solve $|x + 5| = 2x - 1$, rejecting any extraneous solution.
- 5 Solve $|4 - x| = 3x$, rejecting any extraneous solution.

Solving modulus inequalities

- 6 Solve $|2x - 1| < 5$.
- 7 Solve $|x - 3| \geq 4$.
- 8 Solve $3 - |x + 2| > -1$.

Modulus and graph transformations

- 9 The graph shows $y = |x^2 - 4x|$. Explain why the parts of the original parabola $y = x^2 - 4x$ between $x = 0$ and $x = 4$ appear reflected upward.



- 10 Explain the difference between the graphs of $y = |f(x)|$ and $y = f(|x|)$, using $f(x) = x - 2$ as an example.

Finding inverse functions

- 11 Find $f^{-1}(x)$ for $f(x) = 3x + 2$.
- 12 Find $f^{-1}(x)$ for $f(x) = \frac{2x+1}{x-3}$, $x \neq 3$.

Restricting the domain for invertibility

- 13 Explain why $g(x) = x^2 - 6x + 5$ (domain all reals) is not one-to-one, and state a suitable restricted domain that makes g invertible.
- 14 Using the restricted domain $x \geq 3$ from the previous problem, find $g^{-1}(x)$ for $g(x) = (x - 3)^2 - 4$, and state its domain.
- 15 For $h(x) = (x + 2)^2$ restricted to $x \leq -2$, find $h^{-1}(x)$ and state its domain.

Composite functions with modulus

- 16 Given $f(x) = |x - 1|$ and $g(x) = 2x + 3$, find $fg(x)$ in simplified form, and evaluate $fg(0)$.

- 17 Using the same $f(x) = |x - 1|$ and $g(x) = 2x + 3$, find $gf(x)$, and evaluate $gf(-4)$.
- 18 Using $fg(x) = 2|x + 1|$ found earlier, solve $fg(x) = 5$.
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Combining modulus with quadratics and other functions

- 19 Solve $|x^2 - 9| = 7$.
- 20 Solve $|2x - 5| = |x + 1|$ by squaring both sides.
- 21 Solve $|2x + 1| = x + 4$, checking both solutions are valid.
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A well-designed word problem: calibrating a sensor

- 22 This question has three parts. An engineer models the deviation of a sensor reading from its calibrated target as $d(x) = |3x - 12|$, where x is the applied voltage (volts) and $d(x)$ is the deviation (millivolts). (a) Find the voltage at which the deviation is zero. (b) The sensor is "within tolerance" when $d(x) \leq 6$. Solve this inequality to find the acceptable range of voltages. (c) For $x \geq 4$, define the calibration function $c(x) = d(x) + 2x - 5$. Explain why c is invertible on this restricted domain, and find $c^{-1}(x)$.